Limits

Informal definition

f has limit L at a, if $f(x)$ approaches L when x approaches a

$$\lim_{x \to a} f(x) = L$$

Example:

$$\lim_{x \to 5} \frac{x^2 + x - 2}{x - 1} = 7$$

$$\lim_{x \to 1} \frac{x^2 + x - 2}{x - 1} = \frac{(x-1)(x+2)}{x-1} = \lim_{x \to 1} (x+2) = 3$$

we can cancel out $x-1$ because by definition, $x \neq 1$, so $x - 1 \neq 0$

Formal definition

$$\lim_{x \to x_0} f(x) = L$$

$$\iff \forall \varepsilon > 0 \ \exists \delta > 0 : 0 < |x - x_0| < \delta$$

$$\implies |f(x) - L| < \varepsilon$$

Left-Right limits

$$\lim_{x \to a^-} f(x) = \lim_{x \uparrow a} f(x)$$

$$\lim_{x \to a^+} f(x) = \lim_{x \downarrow a} f(x)$$

If these do not match, then $\lim_{x \to a} f(x)$ does not exist

Theorems

$$\lim_{x \to a} f(x) = L, \quad \lim_{x \to a} g(x) = M$$

- $\lim_{x \to a} [f(x) \pm g(x)] = L \pm M$
- $\lim_{x \to a} f(x) \cdot g(x) = L \cdot M$
- $\lim_{x \to a} \frac{f(x)}{g(x)} = \frac{L}{M}, \ M \neq 0$
- $\lim_{x \to a} \sqrt{f(x)} = \sqrt{L}, \ L \geq 0$

Interesting limits

- $$\lim_{x\to 1} \frac{\sqrt{x}-1}{x-1} = \lim_{x\to 1} \frac{(\sqrt{x}-1)(\sqrt{x}+1)}{(x-1)(\sqrt{x}+1)}$$
$$= \lim_{x\to 1} \frac{x-1}{(x-1)(\sqrt{x}+1)}$$
$$= \lim_{x\to 1} \frac{1}{\sqrt{x}+1} = \frac{1}{2}$$

- $$\lim_{x\to 1} |x-1| = 0$$

- $$\lim_{x\to 1} \frac{|x-1|}{x-1}$$

$$\hookrightarrow \left.\begin{aligned} \lim_{x\to 1^+} \frac{|x-1|}{x-1} &= \frac{(x-1)}{x-1} = 1 \\ \lim_{x\to 1^-} \frac{|x-1|}{x-1} &= \frac{-(x-1)}{x-1} = -1 \end{aligned}\right] \text{limit doesn't exist}$$

- $$\lim_{x\to 0} x^2 \cos\frac{1}{x} = 0 \leftarrow$$ Squeeze Theorem

$f(x) = -x^2 \quad g(x) = x^2\cos\left(\frac{1}{x}\right)$

$h(x) = x^2 \quad f(x) \le g(x) \le h(x) \Rightarrow \lim_{x\to 0} g(x) = 0$

Squeeze Theorem

If $f(x) \le g(x) \le h(x)$ on (a, b), $x_0 \in (a,b)$

and $\lim_{x\to x_0} f(x) = \lim_{x\to x_0} h(x) = L$

$\Rightarrow \lim_{x\to x_0} g(x) = L$

Limits at ∞

$$\lim_{x \to \infty} f(x) = L$$

$$\iff \forall \varepsilon > 0 \ \exists M > 0 : x > M$$

$x \in \text{dom}$

$$\implies |f(x) - L| < \varepsilon$$

Examples

$$\lim_{x \to \pm\infty} \frac{x}{\sqrt{x^2+1}}$$

$$= \lim_{x \to \pm\infty} \frac{x}{\sqrt{x^2\left(1 + \frac{1}{x^2}\right)}}$$

$$= \lim_{x \to \pm\infty} \frac{x^{+\infty}}{|x|\sqrt{1 + \cancel{\frac{1}{x^2}}^{\,0}}}$$

$\hookrightarrow$ $+\infty$

Note:

$|x| = \sqrt{x^2}$

$$\sqrt{x^2(a+b)} = \sqrt{x^2} \cdot \sqrt{a+b} = |x|\sqrt{a+b}$$

Asymtotes

(1) Vertical

(2) Horizontal

(3) Oblique

Rational functions

$\frac{P(x) \to \text{deg. } m}{Q(x) \to \text{deg. } n}$

if $m \leq n$, there always is a H.A. of $y = 0$

Analysing complex polinomials

$$f(x) = \frac{x^3 + 4x - 5}{x^2 - 3x + 1} \longrightarrow$$

$$\begin{array}{l|l} x^3 + 4x - 5 & x^2 - 3x + 1 \\ \hline x^3 - 3x^2 + x & x + 3 \\ \hline 3x^2 + 3x - 5 & \\ 3x^2 - 9x + 3 & \\ \hline 12x - 8 & \end{array}$$

$$f(x) = x + 3 + \frac{12 - 8}{x^2 - 3x + 1}$$

Completing the square

$$\sqrt{x^2 + 2x} = \sqrt{x^2 + 2x + 1 - 1}$$

$$= \sqrt{(x+1)^2 - 1}$$

$$= \sqrt{(x+1)^2\left(1 - \frac{1}{(x+1)^2}\right)} = |x+1|\sqrt{1 - \frac{1}{(x+1)^2}}$$

Limits using the definition

Prove that $\lim_{x \to 3} (x^2 + x - 6) = 6$

Proof:

- Let $\varepsilon > 0$
- take $\delta = \min(1,$
- Assume that $0 < |x - \varepsilon| < \delta$

  distance between x and ε is less than δ
- then $|x^2 + x - 6 - 6| = |x^2 + x - 12|$

$$= |(x-3)(x+4)| = |x-3||x+4|$$

$$< \delta |x+4| < 8\delta \leq 6 \frac{\varepsilon}{\delta} = \varepsilon \quad \square$$